

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Code of Conduct:

I M. Farah Bin / 19251434 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Farah Bin

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5 10marks
Problem Understanding	20%	1.6	1.6	1.6	1.6	1.6
Method Selection	20%	1.6	1.6	1.6	1.6	1.6
Solution Process	25%	2.0	2.0	2.0	2.0	2.0
Accuracy of Calculation	20%	1.6	1.6	1.6	1.6	1.6
Interpretation of Results	15%	1.2	1.2	1.2	1.2	1.2

Total Marks

140

38/100

Signature of the Faculty

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Co-2 Assessment Tool-1

Analytical Problem Solving

Given network :

- * Network Address : 192.168.20.0/24
- * Total IP addresses = $2^{32-24} = 256$ addresses (192.168.20.0 to 192.168.20.255)
- * The subnets are allocated from the beginning of the network in order :

Lab A $\rightarrow$ /26

Lab B $\rightarrow$ /27

Lab C $\rightarrow$ /28

Lab A (/26)

* Subnet Mask : 255.255.255.192

* Total addresses : $2^{32-26} = 64$

* Usable hosts : $64 - 2 = 62$

Address details

Network address : 192.168.20.0

First host : 192.168.20.1

Last host : 192.168.20.62

Broadcast Address : 192.168.20.63

Step 2: Lab B (127)

The next available address after Lab A is 192.168.20.64

* Subnet Mask: 255.255.255.224

Total address: $2^{32-27} = 32$

Usable host: $32 - 2 = 30$

Address details:

* Network Address: 192.168.20.64

* First host: 192.168.20.65

* Last host: 192.168.20.94

* Broadcast Address: 192.168.20.95

Step 3: Lab C (28)

The next available address after Lab B is 192.168.20.96

* Subnet mask: 255.255.255.240

Total address = $2^{32-28} = 16$

Usable hosts = $16 - 2 = 14$

Address details:

Network Address: 192.168.20.96

First Host: 192.168.20.97

Last Host: 192.168.20.110

Broadcast Address: 192.168.20.111

System 1:

- IP address : 192.168.1.130
- Subnet mask : 255.255.255.192 (126)

System 2:

- IP address : 192.168.1.190
- Subnet mask : 255.255.255.192 (126)

Determine the subnet size

Subnet Mask = 126

Total IP addresses per subnet = $2^{32-26} = 64$

usable host addresses = $64 - 2 = 62$

The subnet ~~increments~~ increments are 64, so the subnet ranges are:

192.168.1.0/26	192.168.1.0 - 192.168.1.63
192.168.1.64/26	192.168.1.64 - 192.168.1.127
192.168.1.128/26	192.168.1.128 - 192.168.1.191
192.168.1.192/26	192.168.1.192 - 192.168.1.255

System 1 Analysis

IP address : 192.168.1.130

This IP falls within :

192.168.1.128 - 192.168.1.191

Therefore,

Therefore.

- * Network Address : 192.168.1.128
- * Broadcast Address : 192.168.1.191
- * Valid host range : 192.168.1.129 - 192.168.1.190
- * Given IP (192.168.1.190)
- * Usable hosts : 62

Q) Given :

- * Network block : 172.16.0.0/16
- * Departments and subnet masks :

- * HR $\rightarrow$ /25
- * Sales $\rightarrow$ /26
- * IT $\rightarrow$ /27
- * Admin $\rightarrow$ /28

The subnets are allocated sequentially from the beginning of the network

HR Department (/25)

- * Subnet mask : 255.255.255.128
- * Total IP addresses = $2^{32-25} = 128$
- * Usable hosts = $128 - 2 = 126$

Subnet details

Network Address : 172.16.0.0

host: 172.16.0.1

host: 172.16.0.126

Broadcast Address: 172.16.0.127

Sales department (126)

* Next available address: 172.16.0.128

* Subnet Mask: 255.255.255.192

* Total IP addresses = 64

* Usable hosts = 62

Subnet details:

* Network Address: 172.16.0.128

* First host: 172.16.0.129

* Last host: 172.16.0.190

* Broadcast Address: 172.16.0.191

IT department (127)

Next available address: 172.16.0.192

Subnet mask: 255.255.255.224

Total IP addresses = 32

Usable host = 30

Subnet details:

Network Address: 172.16.0.192

First host: 172.16.0.193

Last host: 172.16.0.222

Broadcast Address: 172.16.0.223

Admin department (128)

* Next available address : 172.16.0.224

* Subnet mask : 255.255.255.240

* Total IP addresses = 16

* Usable hosts

Subnet details :

Network Address : 172.16.0.224

First host : 172.16.0.225

Last host : 172.16.0.238

Broadcast Address : 172.16.0.239

Network Address : 172.16.10.0/24

Subnet Mask : /27 (255.255.255.254)

Calculate the total numbers of subnets

original network = /24

new subnet mask = /27

Number of borrowed bits = $27 - 24 = 3$

Number of subnets = $2^3 = 8$

Calculate the number of usable hosts per subnet

Host bits = $32 - 27 = 5$

Total IP addresses per subnet : $2^5 = 32$

usable host addresses : $32 - 2 = 30$

List of 8 subnets

Subnet

1	172.16.10.0	172.16.10.1 - 172.16.10.30	172.16.10.31
2	172.16.10.32	172.16.10.33 - 172.16.10.62	172.16.10.63
3	172.16.10.64	172.16.10.65 - 172.16.10.94	172.16.10.95
4	172.16.10.96	172.16.10.97 - 172.16.10.126	172.16.10.127
5	172.16.10.128	172.16.10.129 - 172.16.10.158	172.16.10.159
6	172.16.10.160	172.16.10.161 - 172.16.10.190	172.16.10.191
7	172.16.10.192	172.16.10.193 - 172.16.10.222	172.16.10.223
8	172.16.10.224	172.16.10.225 - 172.16.10.254	172.16.10.255

Determine the subnet for IP address 172.16.10.78

The IP 172.16.10.78 falls between:

172.16.10.64 - 172.16.10.95

Therefore,

- Subnet: 172.16.10.64/27
- Valid host range: 172.16.10.65 - 172.16.10.94
- Broadcast Address: 172.16.10.95

172.16.10.78 belongs to the subnet 172.16.10.64/27.

Check whether 172.16.10.95 is in the same subnet

The subnet is

- Network Address: 172.16.10.64
- Host Range: 172.16.10.65 - 172.16.10.94
- Broadcast Address: 172.16.10.95

Therefore:

* It belongs to the same subnet

* It cannot be assigned to a host because broadcast addresses are reserved

v6 Address: 2001:0db8:0000:0000:0000:0000:0042:8329

Prefix length: /64

Compress the IPv6 Address

- 1) Remove leading zeros in each hexadecimal block
- 2) Replace one consecutive sequence of all-zero block with :: (only once).

Original Address

2001:0db8:0000:0000:0000:0000:0042:8329

After removing leading zeros

2001:db8:0:0:0:0:42:8329

Compress consecutive zero blocks

2001:db8::42:8329

Compressed IPv6 Address:

2001:db8::42:8329

ii) Expand the Compressed Address

Compressed Address = 2001:db8::42:8329

The :: represent three groups of 0000

Expanded form:

2001:0db8:0000:0000:0000:0000:0042:8329

Expanded IPv6 Address:

2001:0db8:0000:0000:0000:0000:0042:8329

iii) Determine the network Prefix (164)

A/64 Prefix means:

* First 64 bits (first 4 blocks) = Network Prefix

* Last 64 bits (last 4 blocks) = Interface Identifier (Host Portion)

Given:

2001:0db8:0000:0000:0000:8b00:0042:8329

Network Prefix:

2001:0db8:0000:0000::/64

or in compressed form:

2001:db8::/64

iv) Calculate Total Host Addresses

Prefix = 164

Host bits:

$$128 - 64 = 64$$

Total Possible host addresses: 2^{64}

$$2^{64} = 18,446,744,073,709,551,616$$